

Photosynthesis: A Short Primer

Prepared as a fixture for faithful note capture.

1. Introduction

Photosynthesis is the process by which plants, algae and some bacteria convert light energy into chemical energy. The energy is stored in sugars that power nearly every food chain on Earth.

The process takes place in chloroplasts and has two linked stages: the light reactions and the Calvin cycle.

- Light reactions capture energy from photons.
- The Calvin cycle fixes carbon dioxide into sugar.
- Oxygen is released as a by-product.

2. Light reactions

In the thylakoid membranes, chlorophyll absorbs light and drives electrons through a transport chain. Water is split to replace those electrons, and the energy is stored as ATP and NADPH.



Figure 1. Schematic chloroplast with stacked thylakoids.

3. Calvin cycle

In the stroma, the enzyme rubisco attaches carbon dioxide to a five carbon sugar. ATP and NADPH from the light reactions then reduce the product into three carbon sugars that the plant uses to build glucose.

Input	Output	Location
Carbon dioxide	Glucose	Stroma
ATP	ADP	Stroma
NADPH	NADP+	Stroma

4. Summary

Light energy becomes chemical energy in two stages, and the oxygen we breathe is a side effect of splitting water.

